

Experiment No.: 4

1. Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software/Tools Required: Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.

3. Theory:

Function Types: Demonstrates anonymous functions (assigned directly to variables without a name) and arrow functions (concise ES6 => syntax).

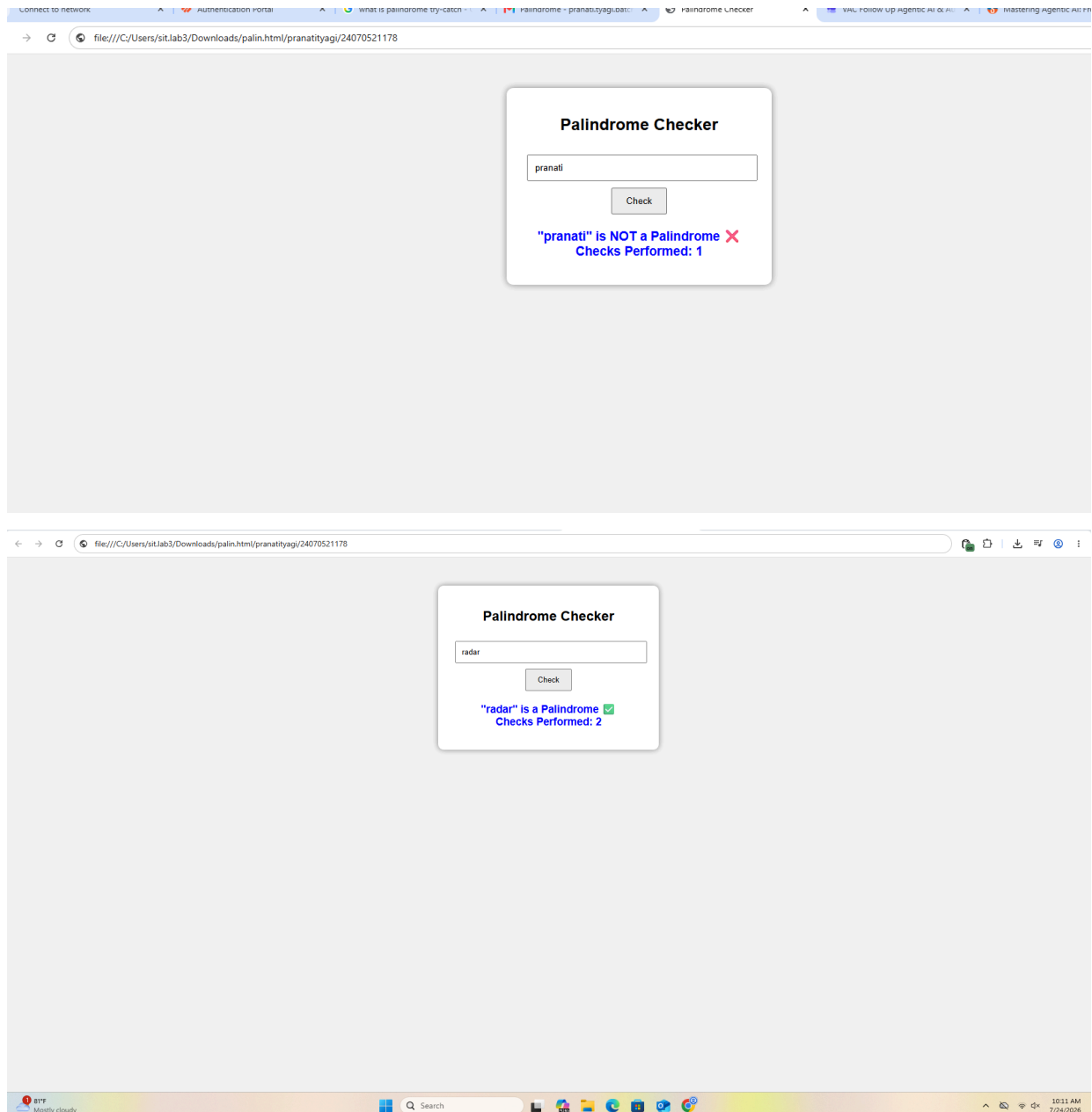
Scope: Differentiates between global variables (accessible throughout the script) and local variables (restricted to the block or function where they are declared).

Closures: Occurs when an inner function retains access to its parent function's scope, preserving state variables (like count) even after the outer function finishes executing.

Error Handling: Uses try...catch and throw statements to validate input types and gracefully handle runtime exceptions without crashing the program.

Palindrome Algorithm: A string is split into an array of characters, reversed, joined back into a string, and compared against the original input using strict equality (===).

4. Output:



5. Result/Conclusion: In this experiment, fundamental JavaScript concepts—including anonymous/arrow functions, scope rules, closures, and try...catch error handling—were successfully implemented. Using these techniques alongside array manipulation methods (split, reverse, join), a functional and error-resistant palindrome checker was constructed.

